Lab1_Q1:

Time and Space complexity
Time: O(1)
Space: O(1)

Lab1_Q2:

Time and Space complexity
Time: O(n)
Space: O(n)

Test 1-
  input:
    ")(",
    "((())",
    "()",
    "(a)((b)((c)(d)))"
  Results:
    String:)(-> INVALID
    String: ((()) -> INVALID
    String: () -> VALID
    String: (a)((b)((c)(d))) -> VALID
Test 2 -
  input:
    "()",
    "(())",
    "(()())",
    "((a+b)*(c-d))",
    "()()()()",
    "(a(b(c)d)e)",
    ")(",
    "((())",
    "())",
    "((a+b)*c",
    "((())())("
  results:
    String: () -> VALID
    String: (()) -> VALID
    String: (()()) -> VALID
    String: ((a+b)*(c-d)) -> VALID
    String: ()()()() -> VALID

String: (a(b(c)d)e) -> VALID
String:)(-> INVALID
String: ((()) -> INVALID
String: ()) -> INVALID
String: ((a+b)*c -> INVALID
String: ((())())(-> INVALID

Lab1_Q3:

Time and Space complexity
Time: O(n)
Space: O(n)

Test 1 -
input:
13, 19, 12, 11, 15, 19, 23, 12, 13, 22, 18, 19, 14, 17, 23, 21
output:
smallest: 11

Test 2 -
input:
42, 7, 89, 56, 23, 95, 11, 68, 34, 77, 2, 61, 50, 14, 83, 29
output:
smallest: 2

Test 3 -
input:
5, 92, 38, 71, 16, 84, 47, 60, 13, 99, 28, 54, 3, 79, 41, 65
output:
smallest: 3

Lab1_Q4:

Time and Space complexity -
Time: O(n^2)
Space: O(n)

Test -
input:
tested with 3, 4 and 8 respectivly
output:
(base) coltonzachary@Coltons-MacBook-Air-2 Alg_Adv_Data_Lab % make run
./a.out 3

```
1, 2, 1
(base) coltonzachary@Coltons-MacBook-Air-2 Alg_Adv_Data_Lab % make run
./a.out 4
1, 3, 3, 1
(base) coltonzachary@Coltons-MacBook-Air-2 Alg_Adv_Data_Lab % make run
./a.out 8
1, 7, 21, 35, 35, 21, 7, 1
(base) coltonzachary@Coltons-MacBook-Air-2 Alg_Adv_Data_Lab %
```